

INDIAN KIDS SCREEN TIME ANALYSIS

ABSTRACT

- Analysis of screen time behavior among Indian children
- Focus on age, gender, location, and device usage
- Insights generated using Power BI
- Highlights health impacts of excessive screen exposure
- Supports decision-making for parents and educators

1. INTRODUCTION

- Increasing use of digital devices among children
- Concerns about excessive screen time
- Health risks: eye strain, sleep issues, obesity
- Objective: analyze patterns and trends in screen usage

Objectives:

- Identify high screen usage groups
- Compare educational vs recreational usage
- Analyze device preferences
- Study urban vs rural differences

2. DATASET DESCRIPTION

- Dataset contains children's screen usage data
- Total records: 82

Key Attributes:

- Age
- Gender

- Avg Daily Screen Time
- Primary Device
- Location (Urban/Rural)
- Usage Type (Educational/Recreational)
- Health Impacts

3. DATA CLEANING & PROCESSING

- Removed duplicates
- Handled missing values
- Standardized categorical data
- Converted data types

Feature Engineering:

- Created usage ratio
- Grouped age into categories
- Derived additional insights

4. EXPLORATORY DATA ANALYSIS

Univariate Analysis:

- Screen time distribution
- Device usage frequency
- Age group distribution

Bivariate Analysis:

- Screen time by gender
- Screen time by age
- Device usage by location

Visualizations Used:

- Bar charts
- Line charts
- Pie/Donut charts

5. DATA MODELING

- Star schema used in Power BI

Tables:

- Fact Table: Screen time data
- Dimension Tables: Age, Gender, Device, Location

Relationships:

- One-to-many relationships for filtering

6. DAX FORMULAS

- Average Screen Time
- Total Participants
- Kids Exceeding Limit
- Educational vs Recreational Ratio
- Urban Count

7. DASHBOARD SLIDE OUTLINE

Slide 1: Overview

- Summary of participants and average screen time
- Overall usage trends

Slide 2: Demographic Patterns

- Screen usage by age, gender, and location

Slide 3: Cohort & Segment Analysis

- Behavior across user groups and devices

Slide 4: Habit & Pattern Analysis

- Screen time distribution and trends

Slide 5: Insights Dashboard

- Key findings and conclusions

8. KEY INSIGHTS

- Average screen time is high (~4.2 hrs/day)
- Majority exceed recommended limits
- Smartphones are most used
- Screen time increases with age
- Urban usage slightly higher

Health Impacts:

- Eye strain
- Poor sleep
- Reduced physical activity

9. TOOLS & TECHNOLOGIES

- Python (Pandas, NumPy)
- Power BI

10. ADVANTAGES

- Clear visualization of patterns
- Identifies high-risk groups
- Supports decision-making
- Raises awareness